```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>The Material Part 8 - Learn HTML 5 Modern</title>
</head>
<body>
    <!-- Continuoes The Material Part 7 - Learn HTML5 Modern -->



                <!-- Chapter 36 - HTML 5 - Cache -->
    <h2> Chapter 36 - HTML 5 Cache </h2>


    <p>
        this is our index.html file    <br>
        <b>
            &lt;!DOCTYPE html&gt;   <br>
            &lt;html manifest="index.appcache"&gt;  <br>
            &lt;body&gt;    <br>
              &lt;p&gt;</b> Content <b>&lt;/p&gt;    <br>
            &lt;/body&gt;   <br>
            &lt;/html&gt;   <br>
        </b>


        <br> <!-- Spacing line like enter -->


        <b>Then Output code is:</b> <br>
        <!-- Output code is : -->
```

```
<!DOCTYPE html>
<html menifest="index.appcache">
<body>
   <p>Content</p>
</body>
</html>
```


 <!-- Spacing line like enter -->

then we will create index.appcache file with below code

 <!-- Spacing line like enter -->

CACHE MANIFEST
<small>index.html</small>

 <!-- Spacing line like enter -->

write those files that you want to be cached load index.html then go for online mode and reload the tab

 <!-- Spacing line like enter -->

Note: The two files must be in the same folder in this example
</p>

 <!-- Spacing line like enter -->

<!-- Chapter 37 - HTML Eevent Attribute -->
<h2> Chapter 37 - HTML Event Attribute </h2>

```
<!-- Chapter 37.1 - HTML Form Events-->
<h3>37.1 - HTML Form Events</h3>

<p>Events triggered by actions inside a HTML form (applies to almost all HTML element, but is most used in form element):</p>

<!-- Emmet HTML code is table>(thead>tr>th*2)+(tbody>tr*10>td*2) -->
<table>
    <thead>
        <tr>
            <th><b>Attribute</b></th>
            <th><b>Description</b></th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>onblur</td>
            <td>Fires the moment that the element loses focus</tr>
        </tr>
        <tr>
            <td>onchange</td>
            <td>Fires the moment when the value of the element is changed</tr>
        </tr>
        <tr>
            <td>oncontextmenu</td>
            <td>Script to be run when a context menu is triggered</td>
        </tr>
        <tr>
            <td>onfocus</td>
            <td>Fires the moment when the element gets focus</td>
```

```
        </tr>
        <tr>
          <td>oninput</td>
          <td>Script to be run when a element gets user input</td>
        </tr>
        <tr>
          <td>oninvalid</td>
          <td>Script to be run when an element is invalid</td>
        </tr>
        <tr>
          <td>onreset</td>
          <td>Fires when the Reset button in a form is clicked</td>
        </tr>
        <tr>
          <td>onsearch</td>
          <td>Fires when the user writes something in search field
(for&lt;input="search"&gt;)</td>
        </tr>
        <tr>
          <td>onselect</td>
          <td>Fires after some text has been selected in an element</td>
        </tr>
        <tr>
          <td>onsubmit</td>
          <td>First when a form is submitted</td>
        </tr>
      </tbody>
    </table>

    <br> <!-- Spacing line like enter -->
```

```
<!-- Chapter 37.2 - Keyboard Events -->
<h3> 37.2 - Keyboard Events </h3>

<!-- Emmet HTML Code is: table>(thead>tr>th*2)+(tbody>tr*3>td*2) -->
<table>
    <thead>
        <tr>
            <th><b>Attribute</b></th>
            <th><b>Description</b></th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td>onkeydown</td>
            <td>Fires when a user is pressing a key</td>
        </tr>
        <tr>
            <td>onkeypress</td>
            <td>Fires when a user presses a key</td>
        </tr>
        <tr>
            <td>onkeyup</td>
            <td>Fires when a user releases a key</td>
        </tr>
    </tbody>
</table>

<br><!-- Spacing line like enter -->
```

<!-- Chapter 38 - Character Entities -->

<h2>Chapter 38 - Character Entities</h2>

<!-- Chapter 38.1 - Character Entities in HTML -->

<h3>38.1 - Character Entities in HTML</h3>

<p>

Character Entities are predefined with 'Entitiy Name'-&entity_name;and'Entity Number'-&entity_number;so we need to use either of the two for the required symbol to be rendered on our page.

 <!--Spacing line like enter -->

The list of few character Entities can be found at <u>https://dev.w3.org/html5/html-author/charref</u>

<!-- Spacing line like enter -->

<input type="text" placeholder="🔍 Search"/>

 <!-- Spacing line like enter -->

Then Output code is:

 <!-- Spacing line like enter -->

Which render as

 <!-- Spacing line like enter -->

<input type="text" placeholder="🔍 Search" />

```
    </p>

    <br> <!-- Spacing line like enter -->

    <!-- Chapter 38.2 - Common Special Characters -->
    <h3> 38.2 - Common Special Characters </h3>

    <p>For this purposes, character entities are cheated. These are of the form
<b>&entity_name;</b> or &entity_number;. The following are some of the available HTML
entities.</p>

    <!-- Emmet HTML code is table>(thead>tr>th*4)+(tbody>tr*10>td*4) -->
    <table>
        <thead>
            <tr>
                <th><b>Character</b></th>
                <th><b>Description</b></th>
                <th><b>Entity Name</b></th>
                <th>Entity Number</th>
            </tr>
        </thead>
        <tbody>
            <tr>
                <td>""</td>
                <td>non-breacking space</td>
                <td>& nbsp;</td>
                <td>&# 160;</td>
            </tr>
            <tr>
                <td>"&lt;"</td>
                <td>less than</td>
```

```
    <td>& lt;</td>
    <td>&# 60;</td>
</tr>
<tr>
    <td>"&gt;"</td>
    <td>greater than</td>
    <td>& gt;</td>
    <td>&# 62 ;</td>
</tr>
<tr>
    <td>"&"</td>
    <td>ampersand</td>
    <td>& amp;</td>
    <td>&# 38;</td>
</tr>
<tr>
    <td>"__"</td>
    <td>em dash</td>
    <td>& mdash;</td>
    <td>&# 8212;</td>
</tr>
<tr>
    <td>"_"</td>
    <td>en dash</td>
    <td>& ndash;</td>
    <td>&# 8211;</td>
</tr>
<tr>
    <td>©</td>
    <td>copyright</td>
```

```
            <td>& copy;</td>
            <td>&# 169 ;</td>
        </tr>
        <tr>
            <td>®</td>
            <td>registered trademark</td>
            <td>& reg;</td>
            <td>&# 174;</td>
        </tr>
        <tr>
            <td>"™”</td>
            <td>trademark</td>
            <td>& reg ;</td>
            <td>&# 8482;</td>
        </tr>
        <tr>
            <td>“☎”</td>
            <td>phone</td>
            <td>& phone ;</td>
            <td>&# 9742;</td>
        </tr>
    </tbody>
</table>

<p>
    Thus, to write  <br>
    © <b>2016 Stack Exchange Inc.</b>   <br>

    <br> <!-- Spacing line like enter -->
```

the following HTML code is used:

 <!-- Spacing line like enter -->

& copy; 2016 Stack Exchange Inc.

<!-- Spacing line like enter -->

Then Output code is:

<!-- Spacing line like enter -->

 © 2016 Stack Exchange Inc.

<!-- Spacing line like enter -->

<!-- Chapter 39 - ARIA -->
<h2> Chapter 39 - ARIA </h2>

<!-- Chapter 39.1 - role="presentation" -->
<h3>39.1 - role="presentation"</h3>

An element whose implicit native role semantics will not be mapped to the accessibility API.

 <!-- Spacing line like enter -->

<!-- mengubah attribute awal "role="float:...;" " pada element div satu(1) dan dua(2) di ubah menjadi "class="float:...;" " pada element div satu(1) dan dua(2) dan mengubah attribute awal " role="clear:both;" " pada element div ketiga menjadi

" class="clear:both;" " -->

 <!-- Spacing line like enter -->

<div class="float:left;"> Some Content on The left </div>

<div class="float:right;"> Some content on the right </div>

<div style="presentation" class="clear:both;"> </div> <!-- Only used to clear floats -->

 <!-- Spacing line like enter -->

Then Output Code is:

<!-- mengubah attribute awal "role="float:...;" " pada element div satu(1) dan dua(2) di ubah menjadi "class="float:...;" " pada element div satu(1) dan dua(2) dan mengubah attribute awal " role="clear:both;" " pada element div ketiga menjadi

" class="clear:both;" " -->

Some Content on The left

Some Content on the right

Presentation <!-- Only used to clear floats -->

</p>

 <!-- Spacing line like enter -->

<!-- Chapter 39.2 - role="alert" -->

<h3> 39.2 - role="alert" </h3>

A message with important, and usually time-sensitive, information.

 <!-- Spacing line like enter -->

<div role="alert" aria-live="assertive"> Your session will expire in 60 seconds. </div>

 <!-- Spacing line like enter -->

Note that i've included bot role="alert" and aria-live="assertive" at the same time.

 <!-- Spacing line like enter -->

Source - Heydon Pickering <u> 'Some Practical ARIA examples' </u>

<!-- Chapter 39.3 - role="alert dialog" -->

<h3> 39.3 - role="alert dialog" </h3>

A type of dialog that contains an alert message, where initial focus goes to an element within the dialog.

 <!-- Spacing line like enter -->

<b>

<div role="alertdiaglog">

 <h1> Warning </h1>

 <div role="alert"> Your session will expire in 60 seconds. </div>

 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<h1>Warning</h1>

Your session wll expire in 60 seconds.

 <!-- Spacing line like enter -->

<!-- Chapter 39.4 - role="application" -->

<h3>39.4 - role="application"</h3>

A region declared as a web application, as opposed to a web document. In this example, the application is a simple calculator that might add two number together.

 <!-- Spacing line like enter -->

<div role="application">

 <h1> Calculator </h1>

 <input title="#" type="text" id="num1"> + <input title="#" type="text" id="id=num2">

```
 &lt;span id="result"&gt; &lt;/span&gt;   <br>
&lt;/div&gt;    <br>
</b>

<br><!-- Spacing line like enter -->

<!-- Output code is-->
<b>Then Output code is:</b> <br>

<div role="application">
    <h1>Calculator</h1>
    <input title="#" type="text" id="num1"> + <input title="#" type="text" id="id=num2">
    <span id="result"></span>
</div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.5 - role="article" -->
<h3>39.5 - role="article"</h3>

<p>
    A section of a page that consists of a composition that forms an independent part of a document, page, or site.     <br>

<br> <!-- Spacing line like enter -->

<a href="https://www.w3.org/TR/html-aria/#sec-strong-native-semantics"><u>Setting an ARIA role and/or aria-* attribute that matches the default implicit ARIA semantics is unnecessary and is not recommended as these properties are already set by the browser.</u></a>   <br>
```

```
    <br> <!-- Spacing line like enter -->

    <b>
        &lt;article&gt; <br>
         &lt;h1&gt;</b> My First article <b>&lt;/h1&gt;   <br>
         &lt;p&gt;</b> Lorem ipsum... <b>&lt;/p&gt;   <br>
        &lt;/article&gt;    <br>
        </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>

    <article>
        <h1>My First article</h1>
        <p>Lorem ipsum...</p>
    </article>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39. - role="" -->
<h3> 39.6 - role="benner"</h3>

<p>
    A region that contains mostly site-oriented content, rather than page-specific content.
<br>

    <br> <!-- Spacing line like enter -->
```

```
    <b>
        &lt;div role="banner"&gt;   <br>
         &lt;h1&gt;</b> My Site <b>&lt;/h1&gt;    <br>

        <br>

          &lt;ul&gt;    <br>
          &lt;li a="" href=""&gt;</b> Home <b>&lt;/li&gt; <br>
          &lt;li a="" href="/about"&gt;</b> About <b>&lt;/li&gt;  <br>
          &lt;li a="" href="/contact"&gt;</b> Contact <b>&lt;/li&gt;  <br>
         &lt;/ul&gt;   <br>
        &lt;/div&gt;   <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <b>Then Output code :</b>   <br>

    <!-- Emmet HTML code is : div[role="banner"]>(h1{My Site})>(ul>li[a
href=""]{Home}+li[a href="/about"]{About}+li[a href="/contact"]{Contact}) -->

    <div role="banner">
        <h1>My Site</h1>
        <ul>
            <li a="" href="">Home</li>
            <li a="" href="/about">About</li>
            <li a="" href="/contact">Contact</li>
        </ul>
    </div>
</p>
```

```
<br> <!-- Spacing line like enter -->

<!-- Chapter 39.7 - role="button" -->
<h3> 39.7 - role="button"</h3>

<p>
    An input that allows for user-triggered actions when clicked or pressed. <br>

    <br> <!-- Spacing line like enter -->

    &lt;button role="button"&gt; Add  &lt;/button&gt;   <br>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>  <br>

    <button role="button">Add</button>
</p>

<br><!-- Spacing line like enter -->

<!-- Chapter 39.8 - role="cell" -->
<h3> 39.8 - role="cell"</h3>

<p>
    A cell in a tabular container. <br>

    <br> <!-- Spacing line like enter -->
```

```
<!-- Emmet HTML code is table>(thead{&lt;!--etc--
&gt;})+(tbody>td[role="cell"]{95}+td[role="cell"]{14}+td[role="cell"]{25}) -->

<b>
   &lt;table&gt;   <br>
    &lt;thead&gt; <br>
     &lt;!--etc--&gt;    <br>
    &lt;/thead&gt;    <br>
    &lt;tbody&gt; <br>
     &lt;td role="cell"&gt;</b> 95 <b>&lt;/td&gt; <br>
     &lt;td role="cell"&gt;</b> 14 <b>&lt;/td&gt; <br>
     &lt;td role="cell"&gt;</b> 25 <b>&lt;/td&gt; <br>
    &lt;/tbody>    <br>
   &lt;/table&gt;    <br>
</b>

<!-- Output code is : -->
<b>Then Output code is:</b> <br>

<!-- Emmet HTML code is table>(thead{&lt;!--etc--
&gt;})+(tbody>td[role="cell"]{95}+td[role="cell"]{14}+td[role="cell"]{25}) -->

<table>
   <thead>
      &lt;!--etc--&gt;
   </thead>
   <tbody>
      <td role="cell">95</td>
      <td role="cell">14</td>
      <td role="cell">25</td>
```

```
        </tbody>
    </table>
</p>



<!-- Chapter 39.9 - role="checkbox" -->
<h3> 39.9 - role="checkbox"</h3>

<p&gt;>
    A checkable input that has three possible value: true, false, or mixed.     <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;p&gt;   <br>
         &lt;input title="#" type="checkbox" role="checkbox" aria-checked="false"&gt;</b>  <br>
         I agree to the terms  <br>
        &lt;/p&gt;  <br>
    </b>

    <br> <!-- Spacing like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>

    <p>
        <input title="#" type="checkbox" role="checkbox" aria-checked="false">
        I agree to the terms
    </p>
```

```
</p>

<br><!-- Spacing line like enter -->


<!-- Chapter 39.10 - role="columnheader" -->
<h3> 39.10 - role="columnheader"</h3>

<p>
   A containing header information for a column. <br>

   <br> <!-- Spacing line like enter -->

   <!-- Emmet HTML code is -->
   <b>
      &lt;table&gt;   <br>
       &lt;thead&gt; <br>
        &lt;tr&gt;  <br>
        &lt;th role="columnheader"&gt;</b> Day 1 <b>&lt;/th&gt;  <br>
        &lt;th role="columnheader"&gt;</b> Day 2 <b>&lt;/th&gt;  <br>
        &lt;th role="columnheader"&gt;</b> Day 3 <b>&lt;/th&gt;  <br>
        &lt;/tr&gt;    <br>
       &lt;/thead&gt;    <br>
       &lt;tbody&gt;   <br>
        &lt;!--etc--&gt;    <br>
       &lt;/tbody&gt;    <br>
      &lt;/table&gt;    <br>
   </b>

   <br> <!-- Spacing line like enter -->
```

```
    <!-- Output code is: table>(thead>tr>th*3)+(tbody>tr>td*3) -->
    <b>Then Output code is:</b> <br>

    <table>
      <thead>
        <tr>
          <th role="columnheader">Day 1</th>
          <th role="columnheader">Day 2</th>
          <th role="columnheader">Day 3</th>
        </tr>
      </thead>
      <tbody>
        <!--etc-->
      </tbody>
    </table>
  </p>

  <br> <!-- Spacing line like enter -->

  <!-- Chapter 39.11 - role="combobox" -->
  <h3> 39.11 - role="combobox"</h3>

  <p>
    A presentation of a select; usually similar to a textbox where users can type ahead to select an option, or type to enter arbitary text as a new item in the list.  <br>

    <br> <!-- Spacing line like enter -->

    <b>
      &lt;input type="text" role="combobox" aria-expanded="false"&gt;  <br>
```

 <input title="#"<!-- aria-autocomplete="list--> ">

</div>

 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<input title="#" type="text" role="combobox" aria-expanded="false">

Typically, you would use JavaScript to build the rest of the typehead or list select functionallity.

 <!-- Spacing line like enter -->

Example (materi tambahan and just optional) :

<!--

<div role="combobox" aria-haspopup="listbox">

 <input title="#" type="text" aria-autocomplete="list">

 <ul role="listbox">

  <li role="option"> Option 1

  <li role="option"> Option 2

  <li role="option"> Option 3

</div>

-->

```
    <!-- Emmet HTML code is div[role="combobox" aria-haspopup="listbox"]>(input[title="#" type="text" aria-autocomplete="list"])+(ul[role="listbox"]>li[role="option"]{Option 1}+li[role="option"]{Option 2}+li[role="option"]{Option 3}) -->


    <br> <!-- Spacing line like enter -->


    <!-- Output code is: -->

    <b>Then Output code is:</b> <br>


    <!-- Just optional to be added " aria-expanded="false" arial-control="my-listbox" " -->

    <div role="combobox" aria-haspopup="listbox" aria-expanded="false" arial-control="my-listbox">

        <input title="#" type="text" aria-autocomplete="list">

        <ul role="listbox">

            <li role="option">Option 1</li>

            <li role="option">Option 2</li>

            <li role="option">Option 3</li>

        </ul>

    </div>

</p>




<!-- Chapter 39.12 - role="complenmentary" -->

<h3> 39.12 - role="complenmentary"</h3>


<p>

    A supporting section of the document, designed to be complementary to the main content at a similar level in the DOM hierarchy, but remains meaningful when separated from the main content.  <br>
```

```
    <br> <!-- Spacing line like enter -->

    <!-- Emmet HTML code is div[role="complementary"]>(h1{More Articles})+(ul{!--etc--}) -->
    <b>
        &lt;div role="complementary"&gt;  <br>
         &lt;h1&gt;</b> More Articles <b>&lt;/h1&gt;    <br>
         &lt;ul&gt;</b> &lt;!--etc--&gt;
        <b>&lt;/ul&gt;   <br>
        &lt;/div&gt;    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <div role="complementary">
        <h1>More Articles</h1>
        <ul><!--etc--></ul>
</p>

<br><!-- Spacing line like enter -->

<!-- Chapter 39.13 - role="contentinfo" -->
<h3> 39.13 - role="contentinfo"</h3>

<p>
    A large perceivable region that contains information about the parent document.     <br>

    <br> <!-- Spacing line like enter -->
```

First

<p role="contentinfo">
Author : Albert Einstein
Published : August 15, 1940
</p>

<!-- Output code is: -->
Then Output code is:

Author : Albert Einstein
Published : August 15, 1940

<!-- Spacing line like enter -->

Second

<!-- Emmet HTML code is div[role="contentinfo"]>(h1{About Us})+(p{We are a company that...}) -->

<div role="contentinfo">
 <h1> About Us </h1>
 <p> We are a company that... </p>
</div>

 <!-- Spacing line like enter -->
<!-- Output code is: -->
Then Output code is:

```
    <div role="contentinfo">
      <h1>About Us</h1>
      <p>We are a company that...</p>
    </div>
  </p>

  <br><!-- Spacing line like enter -->

  <!-- Chapter 39.14 - role="definition" -->
  <h3> 39.14 - role="definition"</h3>

  <p>
    A definition of a term or concept.    <br>

    <br> <!-- Spacing line like enter -->

    <b>First</b> <br>

    <br> <!-- Spacing line like enter -->

    &lt;!-- Emmet HTML code is span[role="term" aria-labelledby="def1"]{Love}+(span[id="def1" role="definition"]{an intense feeling of deep affection}) --&gt;   <br>

    <br> <!-- Spacing line like enter -->

    <b>
      &lt;span role="term" aria-labelledby="def1"&gt;</b> Love <b>&lt;/span&gt;   <br>
      &lt;span id="def1" role="definition"&gt;</b> an intense feeling of deep affection <b>&lt;/span&gt;
    </b>   <br>
```


 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

 <!-- Spacing line like enter -->

Love

an intense feeling of deep affection

 <!-- Spacing line like enter -->

Second

 <!-- Spacing line like enter -->

<div role="definition">

 <h1> Definition of ARIA </h1>

 <p> ARIA stands for Accessible Rich Internet Applications. It is a set of attributes that define ways to make web content and web applications more accessible to people with disabilities. </p>

</div>

 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<h1>Definition of ARIA</h1>

ARIA stands for Accessible Rich Internet Applications. It is a set of attributes that define ways to make web content and web applications more accessible to people with disabilities.

 <!-- Spacing line like enter -->

<!-- Chapter 39.15 - role="diaglog" -->

<h3> 39.15 - role="diaglog"</h3>

A dialog is an application window that is designed to interupt the current processing of an application in order to prompt the user to enter information or require a response.

 <!-- Spacing line like enter -->

First

<!-- Emmet HTML code is div[div="dialog"]>(p{Are you sure?})+(button[role="button"]{Yes})+(button[role="button"]{No}) -->

 <!-- Spacing line like enter -->

<div role="dialog">

 <p> Are you sure? </p>

 <button role="button"> Yes </button>

 <button role="button"> No </button>

</div>

 <!-- Spacing line like enter -->

```
<!-- Output code is -->
<b>Then Output code is:</b> <br>

<!-- Emmet HTML code is: div[div="dialog"]>(p{Are you sure?})+(button[role="button"]{Yes})+(button[role="button"]{No}) -->
<div role="dialog">
    <p>Are you sure?</p>
    <button role="button">Yes</button>
    <button role="button">No</button>
</div>   <br>

<br> <!-- Spacing line like enter -->

<b>Second</b>   <br>

<br> <!-- Spacing line like enter -->

<b>
    &lt;div role="dialog"&gt;  <br>
     &lt;h1&gt;</b> Dialog Title <b>&lt;/h1&gt;    <br>
     &lt;p&gt;</b> Dialog content goes here... <b>&lt;/p&gt;   <br>
    &lt;/div&gt;    <br>
</b>

<br> <!-- Spacing line like enter -->

<!-- Output code is: -->
<b>Then Output code is:</b> <br>
<div role="dialog">
    <h1>Dialog Title</h1>
```

```
        <p>Dialog content goes here...</p>
    </div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.16 - role="directory" -->
<h3> 39.16 - role="directory"</h3>

<p>
    A list of references to members of a group, such a static of contents.  <br>

    <br> <!-- Spacing line like enter -->

    <b>First</b>    <br>

    &lt;ul role="directory"&gt;   <br>
        &lt;li&gt;&lt;a href="/chapter1"&gt;Chapter 1&lt;/a&gt;&lt;/li&gt;
        &lt;li&gt;&lt;a href="/chapter2"&gt;Chapter 2&lt;/a&gt;&lt;/li&gt;
        &lt;li&gt;&lt;a href="/chapter3"&gt;Chapter 3&lt;/a&gt;&lt;/li&gt;
    </ul>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>

    <!-- jadi permasalahan nya ada pada role nya, seharusnya role="list" agar tidak terjadi erro pada element dan attribute li atau menambahkan role="directory" pada semua element "li"-->
    <ul role="directory">
```

```
        <li role="directory">
            <a href="/chapter1">Chapter 1</a>
        </li>
        <li role="directory">
            <a href="/chapter2">Chapter 2</a>
        </li>
        <li role="directory">
            <a href="/chapter3">Chapter 3</a>
        </li>
    </ul>

    <b>Second</b>   <br>

    &lt;!-- Emmet HTML code is nav[role="directory"]>(h1{Table of
Contents})+(ul>li>a[href="#Chapter1"]{Chapter 1}+li>a[href="#Chapter2"]{Chapter
2}+li>a[href="#Chapter3"]{Chapter 3}) --&gt;   <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;nav role="directory"&gt;  <br>
         &lt;h1&gt;</b> Table of Contents <b>&lt;/h1&gt;    <br>
         &lt;ul&gt;</b>   <br>
          &lt;li&gt; &lt;a href="#Chapter1"&gt;</b> Chapter 1 <b>&lt;/li&gt;
<br>
          &lt;li&gt; &lt;a href="#Chapter2"&gt;</b> Chapter 2 <b>&lt;/li&gt;
<br>
          &lt;li&gt; &lt;a href="#Chapter3"&gt;</b> Chapter 3 <b>&lt;/li&gt;
<br>
         &lt;/ul&gt;   <br>
        &lt;/nav&gt;    <br>
    </b>
```

```
    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->

    <!-- Emmet HTML code is nav[role="directory"]>(h1{Table of
Contents})+(ul>li>a[href="#Chapter1"]{Chapter 1}+li>a[href="#Chapter2"]{Chapter
2}+li>a[href="#Chapter3"]{Chapter 3}) --&gt; -->

    <b>Then Output code is:</b> <br>
    <nav role="directory">
      <h1>Table of Contents</h1>
      <ul>
        <li><a href="#Chapter1">Chapter 1</a></li>
        <li><a href="#Chapter2">Chapter 2</a></li>
        <li><a href="#Chapter3">Chapter 3</a></li>
      </ul>
    </nav>
  </p>

  <br> <!-- Spacing line like enter -->

  <!-- Chapter 39.17 - role="document" -->
  <h3> 39.17 - role="document"</h3>

  <p>
    A region declared as a web document, as opposed to a web application.     <br>

    <br> <!-- Spacing line like enter -->

    <b>First</b>    <br>
```

<!-- Emmet HTML code is div[role="document"]>(h1{The life of Albert Einstein})+(p{Lorem ipsum...}) -->

<div role="document">

 <h1> The life of Albert Einstein </h1>

 <p> Lorem ipsum... </p>

</div>

 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<h1>The life of Albert Einstein</h1>

Lorem ipsum...

Second

<!-- Emmet HTML code is div[role="document"]>(h1{My Document})+(p{This is a simple document with some text content.}) -->

<div role="document">

 <h1> My Document </h1>

 <p> This is a simple document with some text content. </p>

</div>

 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<h1>My Document</h1>

This is a simple document with some text content.

 <!-- Spacing line like enter -->

<!-- Chapter 39.18 - role="form" -->

<h3> 39.18 - role="form"</h3>

<P>

A landmark region that contains a collection of items and objects that, as a whole, combine to create a form.

 <!-- Spacing line like enter -->

First

 <!-- Spacing line like enter -->

<!-- Emmet HTML code is : form[action=""]>fieldset>(legend{Login Form})+(div>label[for="username"]{Your username}+input[type="text" id="username" aria-describedby="username-tip" required /]+div[role="tooltip" id="username-tip"]{Your username is your email address})+(div>label[for="password"]{Your Password}+input[type="text" id="password" aria-describedby="password-tip" required/]+div[role="tooltip" id="password-tip"]{Was emailed to you when you signed up}) -->

 <!-- Spacing line like enter -->

<form action="">

 <fieldset>

  <legend> Login Form </legend>

  <div>

   <label for="username"> Your username </label>

   <input type="text" id="username" aria-describedby="username-tip" required />

   <div role="tooltip" id="username-tip"> Your username is your email address </div>

  </div>

  <div>

   <label for="password"> Your Password </label>

   <input type="text" id="password" aria-describedby="password-tip" required />

   <div role="tooltip" id="password-tip"> Was emailed to you when you signed up </div>

  </div>

</fieldset>

</form>

 <!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<!-- Emmet HTML code is : form[action=""]>fieldset>(legend{Login Form})+(div>label[for="username"]{Your username}+input[type="text" id="username"

```
aria-describedby="username-tip" required /]+div[role="tooltip" id="username-tip"]{Your username is your email address})+(div>label[for="password"]{Your Password}+input[type="text" id="password" aria-describedby="password-tip" required/]+div[role="tooltip" id="password-tip"]{Was emailed to you when you signed up})
-->


    <form action="">
        <fieldset>
            <legend>Login Form</legend>
            <div>
                <label for="username">Your username</label>
                <input type="text" id="username" aria-describedby="username-tip" required/>
                <div role="tooltip" id="username-tip">Your username is your email address</div> <br>
            </div>
            <div>
                <label for="password">Your Password</label>
                <input type="text" id="password" aria-describedby="password-tip" required/>
                <div role="tooltip" id="password-tip">Was emailed to you when you signed up</div>
            </div>
        </fieldset> <br>


    <B>Second</B>   <br>


    <br> <!-- Spacing line like enter -->


    &lt;!-- Emmet HTML code is form[role="form"]>(h1{Contact Us})+(label[for="name"]{Name:}+input[type="text" id="name" name="name"])+(label[for="email"]{Email:}+input[type="email" id="email" name="email"])+(label[for="message"]{Message:}+textarea[id="message" name="message"])+(button[type="submit" role="button"]{Send}) --&gt;   <br>


    <br> <!-- Spacing line like enter -->
```

```
<b>
    &lt;form role="form"&gt;  <br>
     &lt;h1&gt;</b> Contact Us <b>&lt;/h1&gt;    <br>
     &lt;label for="name"&gt;</b> Name: <b>&lt;/label&gt;   <br>
     &lt;input type="text" id="name" name="name"&gt;   <br>
     &lt;label for="email"&gt;</b> Email: <b>&lt;/label&gt;  <br>
     &lt;input type="email" id="email" name="email"&gt;    <br>
     &lt;label for="message"&gt;</b> Message: <b>&lt;/label&gt;    <br>
     &lt;textarea id="message" name="message"&gt; &lt;/textarea&gt;  <br>
     &lt;button type="submit" role="button"&gt;</b> Send <b>&lt;/button&gt;
<br>
    &lt;/form&gt;    <br>
</b>

<br> <!-- Spacing line like enter -->

<!-- Output code is: -->
<b>Then Output code is:</b> <br>

<!-- Emmet HTML code is form[role="form"]>(h1{Contact
Us})+(label[for="name"]{Name:}+input[type="text" id="name"
name="name"])+(label[for="email"]{Email:}+input[type="email" id="email"
name="email"])+(label[for="message"]{Message:}+textarea[id="message"
name="message"])+(button[type="submit" role="button"]{Send}) -->
<form role="form">
    <h1>Contact Us</h1>
    <label for="name">Name:</label> <br>
    <input type="text" id="name" name="name">   <br>
    <label for="email">Email:</label>   <br>
    <input type="email" id="email" name="email">    <br>
    <label for="message">Message:</label>   <br>
```

```
        <textarea id="message" name="message"></textarea>   <br>
        <button type="submit" role="button">Send</button>   <br>
    </form> <br>
</P>

<p> You would use role=form on non-semantic elements (not recommended, invalid) </p>

<p>
    <b>
        &lt;div role="form"&gt;</b>  <br>
         &lt;input type="email" placeholder="Your email address"&gt;   <br>
         &lt;button&gt;</b> sign up <b>&lt;/button&gt;    <br>
        &lt;/div&gt;    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>

    <div role="form">
        <input type="email" placeholder="Your email address">
        <button>sign up</button>
    </div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.19 - role="grid" -->
<h3> 39.19 - role="grid"</h3>
```

<p>

A grid is an active control which contains cells of tabular data arranged in rows and columns, like a table.

 <!-- Spacing line like enter-->

First

 <!-- Spacing line like enter -->

<!-- Emmet HTML code is: table[role="grid"]>(thead{!--etc--}+tbody{!--etc--})-->

<table role="grid">
 <thead>
  <!--etc-->
 </thead>
 <tbody>
  <!--etc-->
 </tbody>
</table>

 <!-- Spacing line like counter -->

<!-- Output code is: -->
Then Output code is:

<!-- Emmet HTML code is: table[role="grid"]>(thead{!--etc--})+(tbody{!--etc--}) -->

<table role="grid">
<thead>
<!--etc--> etc
</thead>
<tbody>
<!--etc--> etc
</tbody>
</table>

 <!-- Spacing line like enter -->

<!-- Emmet HTML code is table[role="grid"]>(thead>tr>th*3)+(tbody>tr>td*3) -->

 <!-- Spacing line like enter -->

Second

<!-- Emmet HTML code is table[role="grid"]>(thead>tr>th*3)+(tbody>tr>td*3) --
>

 <!-- Spacing line like enter -->

<table role="grid">
 <thead>
  <tr>
  <th role="columnheader"> Day 1 </th>
  <th role="columnheader"> Day 2 </th>
  <th role="columnheader"> Day 3 </th>
  </tr>
 </thead>

```
    &amp;emsp;&amp;lt;tbody&amp;gt;   <br>
    &amp;emsp;&amp;emsp;&amp;lt;tr&amp;gt;  <br>
    &amp;emsp;&amp;emsp;&amp;emsp;&amp;lt;td role="gridcell"&amp;gt;</b> 95 <b>&amp;lt;/td&amp;gt;    <br>
    &amp;emsp;&amp;emsp;&amp;emsp;&amp;lt;td role="gridcell"&amp;gt;</b> 14 <b>&amp;lt;/td&amp;gt;    <br>
    &amp;emsp;&amp;emsp;&amp;emsp;&amp;lt;td role="gridcell"&amp;gt;</b> 35 <b>&amp;lt;/td&amp;gt;    <br>
    &amp;emsp;&amp;emsp;&amp;lt;/tr&amp;gt;    <br>
    &amp;emsp;&amp;lt;/tbody&amp;gt;    <br>
    &amp;lt;/table&amp;gt;    <br>
</b>

<br> <!-- Spacing line like enter -->
<!-- Output code is: table[role="grid"]>(thead>tr>th*3)+(tbody>tr>td*3) -->
<b>Then Output code is:</b> <br>
<table role="grid">
    <thead>
        <tr>
            <th role="columnheader">Day 1</th>
            <th role="columnheader">Day 2</th>
            <th role="columnheader">Day 3</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td role="gridcell">95</td>
            <td role="gridcell">14</td>
            <td role="gridcell">35</td>
        </tr>
    </tbody>
</table>
```

```
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.20 - role="gridcell" -->
<h3> 39.20 - role="gridcell"</h3>

<p>
   A cell in a grid or treegrid.   <br>

   <br> <!-- Spacing line like enter -->

   &lt;!-- Emmet HTML code is: table[role="grid"]>(thead{!----
})+(tbody>tr>td[role="gridcell"]{17}+td[role="gridcell"]{64}+td[role="gridcell"]{18}) --
&gt;    <br>

   <br> <!-- Spacing line like enter -->

   <b>
      &lt;table role="grid"&gt;   <br>
       &lt;thead&gt; <br>
        &lt;!----   <br>
       &lt;/thead&gt;    <br>
       &lt;tbody&gt; <br>
       &lt;tr&gt;    <br>
        &lt;td role="gridcell"&gt;</b> 17 <b>&lt;/td&gt;   <br>
        &lt;td role="gridcell"&gt;</b> 64 <b>&lt;/td&gt;   <br>
        &lt;td role="gridcell"&gt;</b> 18 <b>&lt;/td&gt;   <br>
        &lt;/tr&gt; <br>
       &lt;/tbody&gt;    <br>
      &lt;/table&gt;  <br>
```

```
    </b>

    <br><!-- Spacing line like enter -->

    <!-- Output code is : -->
    <b>Then Output code is:</b>

    <!-- Emmet HTML code is: table[role="grid"]>(thead{!----
})+(tbody>tr>td[role="gridcell"]{17}+td[role="gridcell"]{64}+td[role="gridcell"]{18}) -->

    <table role="grid">
      <thead>
        <!---->
      </thead>
      <tbody>
        <tr>
          <td role="gridcell">17</td>
          <td role="gridcell">64</td>
          <td role="gridcell">18</td>
        </tr>
      </tbody>
    </table>

  </p>

  <br> <!-- Spacing line like enter -->

  <!-- Chapter 39.21 - role="group" -->
  <h3> 39.21 - role="group"</h3>

  <p>
```

A set of user interface objects which are not intended to be included in page summary or table of contents by assistive technologies.

 <!-- Spacing line like enter -->

<div role="group">

 <button role="button"> Previous </button>

 <button role="button"> Next </button>

</div>

<!-- Spacing line like enter -->

<!-- Output code is: -->

Then Output code is:

<!-- Emmet HTML code is:
div[role="group"]>(button[role="button"]{Previous})+(button[role="button"]{Next}) -->

<button role="button">Previous</button>

<button role="button">Next</button>

</p>

 <!-- Spacing line like enter -->

<!-- Chapter 39.22 - role="heading" -->

<h3> 39.22 - role="heading"</h3>

```
<p&gt;>
   A heading for a section of the page.    <br>

   <br> <!-- Spacing line like enter -->

   <b>
      &lt;h1 role="heading"&gt;</b> Introduction <b>&lt;/h1&gt;  <br>
      &lt;p&gt;</b> Lorem ipsum... <b>&lt;/p&gt; <br>
   </b>

   <br> <!-- Spacing line like enter -->

   <!-- Output code is: -->
   <b>Then Output code is:</b>
   <h1 role="heading">introduction</h1>
   <p>Lorem ipsum...</p>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.23 - role="img" -->
<h3> 39.23 - role="img"</h3>

<p>
  A container for a collection of elements that form an image. <br>

   <br> <!-- Spacing line like enter -->

   <b>
      &lt;figure title="#" role="img"&gt;   <br>
```

```
     &lt;img src="albert.jpg" alt="A cute cat."&gt;    <br>
     &lt;figcaption&gt;</b> This is my cat, Albert <b></figcaption> <br>
    &lt;/figure&gt;   <br>
  </b>

  <br> <!-- Spacing line like enter -->

  <!-- Output code is: -->
  <b>Then Output code is:</b>

  <figure title="#" role="img">
    <img src="albert.jpg" alt="A cute cat.">
    <figcaption>This is my cat, Albert</figcaption>
  </figure>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.24 - role="link" -->
<h3> 39.24 - role="link"</h3>

<p>
  An interactive reference to an internal or external resource that, when activated, cause the user agent to navigate to that resource. <br>

  <br> <!-- Spacing line like enter -->

  In the majority of cases setting an ARIA role and/or aria-* attribute that matches the <a href="https://www.w3.org/TR/html-aria/#sec-strong-native-semantics"><u>default implicit ARIA semantices</u></a> is unnecessary and not recommended as these properties are already set by the browser.    <br>
```

```
    <br> <!-- Spacing line like enter -->

    Source: <a href="https://www.w3.org/TR/html5/dom.html#aria-usage-
note"><u>https://www.w3.org/TR/html5/dom.html#aria-usage-note</u></a>
  </p>

  <br> <!-- Spacing line like enter -->

  <!-- Chapter 39.25 - role="list" -->
  <h3> 39.25 - role="list"</h3>

  <p>
    A group of non-interactive list items.     <br>

    <br> <!-- Spacing line like enter -->

    <b>
      &lt;ul role="list"&gt;   <br>
       &lt;li role="listitem"&gt; One &lt;/li&gt;    <br>
       &lt;li role="listitem"&gt; Two &lt;/li&gt;    <br>
       &lt;li role="listitem"&gt; Three &lt;/li&gt;  <br>
      &lt;/ul&gt;   <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>

    <!-- Emmet HTML code is: ul[role="list"]>(li[role="listitem"])*3 -->
    <ul role="list">
```

```
        <li role="listitem">one</li>
        <li role="listitem">Two</li>
        <li role="listitem">Three</li>
    </ul>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.26 - role="listbox" -->
<h3> 39.26 - role="listbox"</h3>

<p>
    A widget that allows the user to select one or more items from a list of choices.   <br>

    <br> <!-- Spacing line like enter -->

    &lt;ul role="listbox"&gt; <br>
     &lt;li&gt; One &lt;/li&gt;    <br>
     &lt;li&gt; Two &lt;/li&gt;    <br>
     &lt;li&gt; Three &lt;/li&gt;  <br>
    </ul>

    <br> <!-- Spacing like enter -->

    <!-- Output code is : -->
    <b> Then Output code is: </b>

    <!-- just optional to be added title on the ul "title="listbox", add role="option" to at all li
"-->
    <ul title="listbox" role="listbox">
        <li role="option">One</li>
```

```
        <li role="option">Two</li>
        <li role="option">Three</li>
    </ul>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.27 - role="listiem" -->
<h3> 39.27 - role="listiem"</h3>

<p>
    A single item in a list or directory.   <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;ul role="list"&gt;  <br>
         &lt;li role="listitem"&gt;</b> One <b>&lt;/li&gt;    <br>
         &lt;li role="listitem"&gt;</b> Two <b>&lt;/li&gt;    <br>
         &lt;li role="listitem"&gt;</b> Three <b>&lt;/li&gt;  <br>
        &lt;/ul&gt; <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>

    <ul role="list">
        <li role="listitem">One</li>
```

```
        <li role="listitem">Two</li>
        <li role="listitem">Three</li>
    </ul>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.28 - role="log" -->
<h3> 39.28 - role="log"</h3>

<p>
    A type of live region where new information is added in meaningful order and old information may disappear.     <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;ul role="log"&gt;   <br>
         &lt;li&gt;</b> User 1 Logged in. <b>&lt;/li&gt;  <br>
         &lt;li&gt;</b> User 2 Logged in. <b>&lt;/li&gt;  <br>
         &lt;li&gt;</b> User 3 Logged in. <b>&lt;/li&gt;  <br>
        &lt;/ul&gt;   <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <!-- just optional to be added " role="log" " at all li elements -->
    <ul role="log">
        <li role="log">User 1 Logged in.</li>
```

```
        <li role="log">User 2 Logged in.</li>
        <li role="log">User 3 Logged in.</li>
    </ul>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.29 - role="main" -->
<h3> 39.29 - role="main"</h3>

<p>
    The main content of a document.     <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;!-- header & nav here --&gt;<br>
        &lt;div role="main"&gt; <br>
         &lt;p&gt; Lorem ipsum ... &gt;/p&gt;   <br>
        &lt;/div&gt;  <br>
        &lt;!-- Footer here --&gt;    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <!-- header & nav here -->
        <div role="main">
            <p>Lorem ipsum...</p>
```

```
        </div>
        <!-- Footer here -->
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.30 - role="marquee" -->
<h3> 39.30 - role="marquee"</h3>

<p>
    A type of live region where non-essential information changers frequently.  <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;ul role="marquee"&gt;   <br>
         &lt;li&gt; Dow +0.26% &lt;/li&gt; <br>
         &lt;li&gt;Nasdaq +0.54% &lt;/li&gt;   <br>
         &lt;li&gt; S& amp;p +0.44% &lt;/li&gt;    <br>
        &lt;/ul&gt; <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <!-- Just optional to be added " role="marquee" to at all li element li "-->
    <ul role="marquee">
        <li role="marquee"> Dow +0.26% </li>
        <li role="marquee"> Nasdaq +0.54% </li>
```

```
        <li role="marquee"> S&amp;amp;p +0.44% </li>
    </ul>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.31 - role="math" -->
<h3> 39.31 - role="math"</h3>

<p>
    Content that represents a mathematical expression.  <br>

    <br> <!-- Spacing line like enter -->

    &lt;img src="slope.png" alt="y=mx+b" role="math"&gt;    <br>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Ouput code is:</b>  <br>
    <img src="slope.png" alt="y=mx+b" role="math">
</p>



<!-- Chapter 39.32 - role="menu" -->
<h3> 39.32 - role="menu"</h3>

<p>
    A type of widget that offers a list of choices to the user.     <br>
```

```
    <br> <!-- Spacing line like enter -->

    <b>
       &lt;ul role="menu"&gt;  <br>
        &lt;li role="menuitem"&gt;</b> New <b>&lt;/li&gt;    <br>
        &lt;li role="menuitem"&gt;</b> Open <b>&lt;/li&gt;   <br>
        &lt;li role="menuitem"&gt;</b> Save <b>&lt;/li&gt;    <br>
        &lt;li role="menuitem&gt;</b> Close <b>&lt;/li&gt;   <br>
       &lt;/ul&gt; <br>
    </b>

    <br><!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <ul role="menu">
       <li role="menuitem">New</li>
       <li role="menuitem">Open</li>
       <li role="menuitem">Save</li>
       <li role="menuitem">Close</li>
    </ul>
  </p>

  <br><!-- Spacing line like enter -->

  <!-- Chapter 39.33 - role="menubar" -->
  <h3> 39.33 - role="menubar"</h3>

  <p>
     A presentation of menu that usually remains visible and is usually presented
horizontally.  <br>
```

```
    <br> <!-- Spacing line like enter -->

    <b>
        &lt;ul role="menubar"&gt;   <br>
         &lt;li role="menuitem"&gt;</b> File <b>&lt;/li&gt;   <br>
         &lt;li role="menuitem"&gt;</b> Edit <b>&lt;/li&gt;   <br>
         &lt;li role="menuitem"&gt;</b> View <b>&lt;/li&gt;   <br>
         &lt;li role="menuitem"&gt;</b> Help <b>&lt;/li&gt;   <br>
        &lt;/li&gt;   <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>

    <ul role="menubar">
        <li role="menuitem">File</li>
        <li role="menuitem">Edit</li>
        <li role="menuitem">View</li>
        <li role="menuitem">Help</li>
    </ul>
</p>

<!-- Chapter 39.34 - role="menuitem" -->
<h3> 39.34 - role="menuitem"</h3>

<p>
    An option in a group of choices contained by a menu or menubar.     <br>
```

```
    <br> <!-- Spacing line like enter -->

    <b>
        &lt;ul role="menubar"&gt;  <br>
         &lt;li role="menuitem"&gt;</b> File <b>&lt;/li&gt;  <br>
         &lt;li role="menuitem"&gt;</b> Edit <b>&lt;/li&gt;  <br>
         &lt;li role="menuitem"&gt;</b> View <b>&lt;/li&gt;  <br>
         &lt;li role="menuitem"&gt;</b> Help <b>&lt;/li&gt;  <br>
        &lt;/li&gt;  <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is</b>

    <ul role="menubar">
        <li role="menuitem">File</li>
        <li role="menuitem">Edit</li>
        <li role="menuitem">View</li>
        <li role="menuitem">Help</li>
    </ul>
</p>


<!-- Chapter 39.35 - role="menuitemcheckbox" -->
<h3> 39.35 - role="menuitemcheckbox"</h3>

<p>
```

A checkable menuitem that has three possible value: true, false, mixed.

 <!-- Spacing line like enter -->

<ul role="menu">

 <li role="menuitem"> Console

 <li role="menuitem"> Layout

 <li role="menuitemcheckbox" aria-checked="true"> Word Wrap

<!-- Spacing line like enter -->

<!-- Output code is -->

Then Output code is:

<ul role="menu">

<li role="menuitem">Console

<li role="menuitem">Layout

<li role="menuitemcheckbox" aria-checked="true">Word Wrap

<!-- Chapter 39.36 - role="menuitemradio" -->

<h3> 39.36 - role="menuitemradio"</h3>

A checkable menuitem in a group of menu item radio roles, only one of which can be checked at a time.

```
    <br> <!-- Spacing line like enter -->


    <b>

      &lt;ul role="menu"&gt;  <br>

       &lt;li role="menuitemradio" aria-checked="true"&gt;</b> Left
<b>&lt;/li&gt;  <br>

       &lt;li role="menuitemradio" aria-checked="false"&gt;</b> Center
<b>&lt;/li&gt;  <br>

       &lt;li role="menuitemradio" aria-checked="false"&gt;</b> Right
<b>&lt;/li&gt;    <br>

      &lt;/li&gt; <br>

    </b>


    <br><!-- Spacing line like -->


    <!-- Output code is: -->

    <b>Then Output code is:</b>

    <ul role="menu">

      <li role="menuitemradio" aria-checked="true">Left</li>

      <li role="menuitemradio" aria-checked="false">Center</li>

      <li role="menuitemradio" aria-checked="false">Right</li>

    </ul>

  </p>



  <!-- Chapter 39.37 - role="navigation" -->

  <h3> 39.37 - role="navigation"</h3>


  <p>

    A collection of navigational elements (usually links) for navigating the document or
related documents. <br>
```

```
    <br> <!-- Spacing line like enter -->


    <b>

      &lt;ul role="navigation"&gt;    <br>

       &lt;li role="navigation"&gt; &lt;a href="/"&gt;</b> Home <b>&lt;/a&gt;
&lt;/li&gt;   <br>

       &lt;li role="navigation"&gt; &lt;a href="/About"&gt;</b> About
<b>&lt;/a&gt; &lt;/li&gt; <br>

       &lt;li role="navigation"&gt; &lt;a href="/contact"&gt;</b> Contact
<b>&lt;/a&gt; &lt;/li&gt;   <br>

      &lt;/li&gt;   <br>

    </b>


    <br> <!-- Spacing line like enter -->


    <!-- Output code is: -->

    <b>Then Output code is:</b>

    <!-- Just optional to added " role="navigation" " to at all li element -->


    <ul role="navigation">

      <li role="navigation"><a href="/">Home</a></li>

      <li role="navigation"><a href="/About">About</a></li>

      <li role="navigation"><a href="/contact">Contact</a></li>

    </ul>

  </p>



  <!-- Chapter 39.38 - role="note" -->

  <h3> 39.38- role="note"</h3>
```

```
  <p>
    A sectin whose content is paranthetic or ancillary to the main content of the resource.
<br>

    <br><!-- Spacing line like enter -->

    <b>
      &lt;p&gt;</b> Lorem ipsum... <b>&lt;/p&gt;    <br>
      &lt;p&gt;</b> Lorem ipsum...<b>&lt;/p&gt;  <br>
      &lt;p role="note"&gt;</b> Lorem ipsum... <b>&lt;/p&gt;  <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <p>Lorem ipsum...</p>
    <p>Lorem ipsum...</p>
    <p role="note">Lorem ipsum...</p>
  </p>

  <br><!-- Spacing line like enter -->

  <!-- Chapter 39.39 - role="option" -->
  <h3> 39.39 - role="option"</h3>

  <p>
    A selectable item in a select list.     <br>

    <br><!-- Spacing line like enter -->
```

```
    <b>
        &lt;ul title="#" role="listbox"&gt; <br>
         &lt;li role="option"&gt; Option 1 &lt;/li&gt; <br>
         &lt;li role="option"&gt; Option 2 &lt;/li&gt; <br>
         &lt;li role="option"&gt; Option 3 &lt;/li&gt; <br>
        &lt;/li&gt; <br>
    </b>

    <br><!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <ul title="#" role="listbox">
        <li role="option">Option 1</li>
        <li role="option">Option 2</li>
        <li role="option">Option 3</li>
    </ul>
</p>



<!-- Chapter 39.40 - role="progresbar" -->
<h3> 39.40 - role="progresbar"</h3>

<p>
    An element that displays the progress status for tasks that take a long time.  <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;progress title="#" role="progressbar" value="25" max="100"&gt; 25
&lt;/progress&gt; <br>
```

```
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is:-->
    <b>Then Output code is:</b> <br>
    <progress title="#" role="progressbar" value="25" max="100">25</progress>
  </p>

  <br><!-- Spacing line like enter -->

  <!-- Chapter 39.41 - role="radio" -->
  <h3> 39.41 - role="radio"</h3>

  <p>
    A checkable input in a group of radio roles, only one of which can be checked at a time.
<br>

    <br> <!-- Spacing line like enter -->

    <b>
      &lt;div role="radiogroup"&gt;   <br>

       &lt;input title="#" type="radio" role="radio" aria-checked="true"&gt;</b>
One <b>&lt;br&gt;  <br>

       &lt;input title="#" type="radio" role="radio" aria-checked="false"&gt;</b>
Two <b>&lt;br&gt;  <br>

       &lt;input title="#" type="radio" role="radio" aria-checked="false"&gt;</b>
Three

      <b>&lt;/div&gt; <br>
```

```
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>

    <div role="radiogroup">
       <input title="#" type="radio" role="radio" aria-checked="true"> One <br>
       <input title="#" type="radio" role="radio" aria-checked="false"> Two <br>
       <input title="#" type="radio" role="radio" aria-checked="false"> Three
    </div>
</p>



<!-- Chapter 39.42 - role="region" -->
<h3> 39.42 - role="region"</h3>

<p>
    A large perceivable section of a web page or document, that the author feels is important enough to be included in a page summary or table of contents, for example, an area of the page containing live sporting event statistics. <br>

    <br> <!-- Spacing line like enter -->

    <b>
       &lt;div role="region"&gt; <br> </b>
        Home team: 4 <b>&lt;br&gt;</b>  <br>
        Away team: 2   <br>
       <b>&lt;/div&gt;</b>  <br>
    </b>
```

```
    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <div role="region">
       Home team: 4 <br>
       Away team: 2
    </div>
  </p>


  <!-- Chapter 39.43 - role="radiogroup" -->
  <h3> 39.43 - role="radiogroup"</h3>

  <p>
    A group of radio buttons.    <br>

    <br> <!-- Spacing line like enter -->

    <b>
       &lt;div role="radiogroup"&gt;   <br>

        &lt;input title="#" type="radio" role="radio" aria-checked="true"&gt;</b>
One <b>&lt;br&gt;  <br>

        &lt;input title="#" type="radio" role="radio" aria-checked="false"&gt;</b>
Two <b>&lt;br&gt;  <br>

        &lt;input title="#" type="radio" role="radio" aria-checked="false"&gt;</b>
Three  <br>
```

```
    <b>&lt;/div&gt; <br>
  </b>

  <br> <!-- Spacing line like enter -->

  <!-- Output code is: -->
  <b>Then Output code is:</b> <br>

  <div role="radiogroup">
    <input title="#" type="radio" role="radio" aria-checked="true"> One <br>
    <input title="#" type="radio" role="radio" aria-checked="false"> Two <br>
    <input title="#" type="radio" role="radio" aria-checked="false"> Three
  </div>
</p>

<!-- Chapter 39.44 - role="row" -->
<h3> 39.44 - role="row"</h3>

<p>
  A row of cells in a tabular container.  <br>

  <br> <!-- Spacing line like enter -->

  <b>
    &lt;table&gt;    <br>
     &lt;thead&gt;</b> &lt;!--etc--&gt; <b>&lt;/thead&gt;   <br>
     &lt;tbody&gt; <br>
      &lt;tr role="row"&gt;</b> &lt;!--etc--&gt; <b>&lt;/tr&gt;   <br>
     &lt;/tbody&gt    <br>
    &lt;/table&gt;    <br>
```

```
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <table>
      <thead><!--etc--></thead>  <br>
      <tbody>
        <tr role="row">
          <td><!--etc--></td>
        </tr>
      </tbody>
    </table>
  </p>

  <br><!-- Spacing line like enter -->

  <!-- Chapter 39.45 - role="rowgroup" -->
  <h3> 39.45 - role="rowgroup"</h3>

  <p>
    A group containing one or more row elements in a grid.  <br>

    <br> <!-- Spacing line like enter -->

    <b>
      &lt;table&gt;    <br>
       &lt;thead role="rowgroup"&gt;</b> &lt;!--etc--&gt; <b>&lt;/thead&gt;
<br>
```

```
        &amp;emsp;&amp;lt;tbody role="rowgroup"&amp;gt;</b> &amp;lt;!--etc--&amp;gt; <b>&amp;lt;/tbody&amp;gt;
<br>
        &amp;lt;/table&amp;gt;    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <table>
        <thead><!--etc-->row - 1st</thead>   <br>
        <tbody row="rowgroup"> <!--etc-->row - 2nd</tbody>
    </table>
    </b>
  </p>

  <br> <!-- Spacing line like enter -->

  <!-- Chapter 39.46 - role="rowheader" -->
  <h3> 39.46 - role="rowheader"</h3>

  <p>
    A cell containing header information for a row in a grid.   <br>

    <br> <!-- Spacing line like enter -->

    &amp;lt;!-- Emmet HTML code is table[role="grid"]>(thead{&amp;lt;!--etc--
&amp;gt;})+(tbody>tr>th[role="rowheader"]{Day 1}+td{65})+(tr>th[role="rowheader"]{Day
2}+td{74}) --&amp;gt;   <br>

    <br><!-- Spacing line like enter -->
```

```
<b>
    &lt;table role="grid"&gt;   <br>
     &lt;thead&gt; <!--etc--> &lt;/thead&gt;   <br>
     &lt;tbody&gt; <br>
      &lt;tr&gt;    <br>
       &lt;th role="rowheader"&gt;</b> Day 1 <b>&lt;/th&gt; <br>
       &lt;td&gt;</b> 65 <b>&lt;/td&gt; <br>
      &lt;/tr&gt;   <br>
     &lt;/tbody&gt;    <br>
     &lt;tr&gt;    <br>
       &lt;th role="rowheader"&gt;</b> Day 2 <b>&lt;/th&gt; <br>
       &lt;td&gt;</b> 74 <b>&lt;/td&gt; <br>
      &lt;tr&gt;   <br>
    &lt;/table&gt;    <br>
</b>

<br> <!-- Spacing line like enter -->

<!-- Output code is : -->
<b>Then Output code is:</b> <br>

<!-- Emmet HTML code is table[role="grid"]>(thead{!--etc--
})+(tbody>tr>th[role="rowheader"]{Day 1}+td{65})+(tr>th[role="rowheader"]{Day
2}+td{74}) -->

<table role="grid">
    <thead><!--etc--></thead>
    <tbody>
        <tr>
            <th role="rowheader">Day 1</th>
```

```
            <td>65</td>
          </tr>
        </tbody>
        <tr>
          <th role="rowheader">Day 2</th>
          <td>74</td>
        </tr>
      </table>

  </p>



  <!-- Chapter 39.47 - role="scollbar" -->
  <h3> 39.47 - role="scollbar"</h3>


  <p>
      A graphical object that controls the scrolling of content within a viewing area, regardless of wheter the content is fully display within the viewing area.     <br>


      <br> <!-- Spacing line like enter -->


      <!-- Emmet HTML code is: (div[ide="conten1"]{Lorem ipsum ...})+div[role="scollbar" aria-control="content1" aria-orientation="vertical" aria-valuemax="100" aria-valuemin="0" aria-valuenow="25"]>div[class="srollhandle"] -->


      <b>
        &lt;div id="conten1"&gt;</b> Lorem ipsum ... <b>&lt;/div&gt;  <br>
        &lt; div    <br>
         role="scrollbar"   <br>
         aria-controls="content1"   <br>
         aria-orientation="vertical"   <br>
```

```
         aria-valuemax="100"   <br>
         aria-valuemin="0"     <br>
         aria-label="Scrollbar"    <br>
         aria-valuenow="25"&gt;   <br>
          &lt;div class="scrollhandle"&gt; &lt;/div&gt;    <br>
        &lt;/div&gt;    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <!-- Emmet HTML code is: (div[ide="conten1"]{Lorem ipsum ...})+div[role="scollbar" aria-control="content1" aria-orientation="vertical" aria-valuemax="100" aria-valuemin="0" aria-valuenow="25"]>div[class="srollhandle"] -->

    <div id="content1">Lorem ipsum ...</div>
    <div
        role="scrollbar"
        aria-controls="content1"
        aria-orientation="vertical"
        aria-valuemax="100"
        aria-valuemin="0"
        aria-valuenow="25"
        aria-label="Scrollbar">
            <div class="scrollhandle"></div>
    </div>
</p>

<br> <!-- Spacing line like enter -->
```

```
<!-- Chapter 39.48 - role="searc" -->
<h3> 39.48 - role="search"</h3>

<p>
    A landmark region that contains a collection of items and object that, as a whole, combine to create a search facility.  <br>

    <br> <!-- Spacing line like enter -->

    <!-- Emmet HTML code is : div[role="search"]>input[role="searchbox" type="text"]+(button[role="button"]{searh})-->
    <b>
        &lt;div role="search"&gt;  <br>
         &lt;input title="#" type="text" role="searchbox"&gt;  <br>
         &lt;button role="button"&gt;</b> Searh <b>&lt;/button&gt;  <br>
        &lt;/div&gt;  <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b>
    <div role="search">
        <input title="#" type="text" role="searchbox">
        <button role="button">Searh</button>
    </div>
</p>

<br> <!-- Spacing line like enter -->
```

```
<!-- Chapter 39.49 - role="searchbox" -->
<h3> 39.49 - role="searchbox"</h3>

<p>
   A type pf textbox intended for specifying search criteria.  <br>

   <br><!-- Spacing line like enter -->

   <b>
      &lt;div role="search"&gt;    <br>
       &lt;input title="#" type="text" role="searchbox"&gt;  <br>
       &lt;button role="button"&gt;</b> Searh <b>&lt;/button&gt;   <br>
      &lt;/div&gt;    <br>
   </b>

   <br> <!-- Spacing line like enter -->

   <!-- Output code is: -->
   <b>Then Output code is:</b>
   <div role="search">
      <input title="#" type="text" role="searchbox">
      <button role="button">Searh</button>
   </div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.50 - role="separator" -->
<h3> 39.50 - role="separator"</h3>
```

```
  <p>
    A divider that separates and distinguishes section of content or groups of menitems.
<br>

    <br> <!-- Spacing line like enter -->

    <b>
      &lt;p&gt;</b> Lorem ipsum... <b>&lt;/p&gt;   <br>
      &lt;hr role="separator"&gt;   <br>
      &lt;p&gt;</b> Lorem ipsum... <b>&lt;/p&gt;</b>    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is : -->
    <b>Then Output code is:</b> <br>

    <p>Lorem ipsum...</p>
    <hr role="separator">
    <p>Lorem ipsum...</p>
  </p>

  <br> <!-- Spacing line like enter -->

  <!-- Chapter 39.51 - role="slider" -->
  <h3> 39.51 - role="slider"</h3>

  <p>
    <b>
      &lt;div     <br>
       role="slider" <br>
```

```
        aria-voluemax="100"    <br>
        aria-valuemin="0"   <br>
        aria-valuenow="25"&gt; <br>
      &lt;div class="sliderhandle"&lt; &lt;/div&gt;
    &lt;/div&gt;    <br>
  </b>

  <br> <!-- Spacing line like enter -->

  <!-- Output code is: -->
  <b>Then Output code is:</b>

  <!-- Just added aria-label="slider" dan tabindex="0" -->
  <div role="slider"
    aria-valuemax="100"
    aria-valuemin="0"
    aria-label="slider"
    tabindex="0"
    aria-valuenow="25">
      <div class="sliderhandle"></div>
  </div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.52 - role="spinbutton" -->
<h3> 39.52 - role="spinbutton"</h3>

<p>
  A form of range that expects the user to select from among discrete choices.    <br>
```

```
    <br> <!-- Spacing line like enter -->

    <b>
        &lt;input   <br>
         type="number" <br>
         role="spinbutton"     <br>
         aria-valuemax="100"   <br>
         aria-valuemin="0"     <br>
         aria-valuenow="25"    <br>
         value="25"&gt;    <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <!-- Just added aria="spinbutton"-->
    <input
        type="number"
        role="spinbutton"
        aria-valuemax="100"
        aria-valuemin="0"
        aria-valuenow="25"
        aria-label="spinbutton"
        value="25">
</p>

<br><!-- Spacing line like enter -->
```

<!-- Chapter 39.53 - role="status" -->

<h3> 39.53 - role="status"</h3>

A container whose content is advisory information for the user but is not importan enough to justify an alert, often but not necessary presented as a status bar.

 <!-- Spacing line like enter -->

<div role="status"> Online </div>

<!-- Output code is: -->

Then Output code is:

Online

 <!-- Spacing line like enter -->

<!-- Chapter 39.54 - role="switch" -->

<h3> 39.54 - role="switch"</h3>

A type of checkbox that represents on/off values, as opposed to checked/unchecked values.

<!-- Spacing line like enter -->

<select title="#" role="switch" aria-checked="false">

```
     &lt;option&gt;</b> On <b>&lt;/option&gt; <br>
     &lt;option selected&gt;</b> Off <b>&lt;/option&gt;  <br>
    &lt;/select&gt; <br>
  </b>

  <br> <!-- Spacing line like enter -->

  <!-- Output code is: -->
  <b>Then Output code</b> <br>
  <select title="#" role="switch" aria-checked="false">
    <option> On </option>
    <option selected>Off</option>
  </select>
</p>

<br><!-- Spacing line like enter -->

<!-- Chapter 39.55 - role="tab" -->
<h3> 39.55 - role="tab"</h3>

<p>
  A grouping label providing a mechanism for selecting the tab content that is to be rendered to the user.  <br>

  <br><!-- Spacing line like enter -->

  <b>
    &lt;ul role="tablist"&gt; <br>
     &lt;li role="tab"&gt;</b> Introduction <b>&lt;/li&gt;  <br>
     &lt;li role="tab"&gt;</b> Chapter 1 <b>&lt;/li&gt;  <br>
     &lt;li role="tab"&gt;</b> Chapter 2 <b>&lt;/li&gt;  <br>
```

```
    &lt;/li&gt;   <br>
  </b>

  <br> <!-- Spacing line like enter -->

  <!-- Output code is:-->
  <b>Then Output code is:</b>
  <ul role="tablist">
    <li role="tab">Introduction</li>
    <li role="tab">Chapter 1</li>
    <li role="tab">Chapter 2</li>
  </ul>
</p>


<!-- Chapter 39.56 - role="table" -->
<h3> 39.56 - role="table"</h3>

<p>
  A section containing data arranged in rows and columns. The table role is intended for tabular containers which are not interactive.    <br>

  <br> <!-- Spacing line like enter -->

  <b>
    &lt;table role="table"&gt;  <br>
     &lt;thead&gt; &lt;!----&gt; &lt;/thead&gt;  <br>
     &lt;tbody> &lt;!----&gt; &lt;/tbody&gt;   <br>
    &lt;/table&gt;  <br>
  </b>
```

```
    <br> <!-- Spacing line like enter -->

    <!-- Spacing line like enter -->
    <b>Then Output code is:</b>
    <table role="table">
        <thead><!----></thead>
        <tbody><!----></tbody>
    </table>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.57 - role="tablist" -->
<h3> 39.57 - role="tablist"</h3>

<p>
    A list of tab elements, which are references to tabpanel elements.  <br>

    <br> <!-- Spacing like enter -->

    <b>
        &lt;ul role="tablist"&gt; <br>
         &lt;li role="tab"&gt;</b> Introduction <b>&lt;/li&gt;    <br>
         &lt;li role="tab"&gt;</b> Chapter 1 <b>&lt;/li&gt;   <br>
         &lt;li role="tab"&gt;</b> Chapter 2 <b>&lt;/li&gt;   <br>
        &lt;/li&gt;   <br>
    </b>

    <br> <!-- Spacing line like enter -->
```

```
    <!-- Output code is:-->
    <b>Then Output code is:</b>
    <ul role="tablist">
       <li role="tab">Introduction</li>
       <li role="tab">Chapter 1</li>
       <li role="tab">Chapter 2</li>
    </ul>
  </p>



  <!-- Chapter 39.58 - role="tabpanel" -->
  <h3> 39.58 - role="tabpanel"</h3>


  <p>
     A container for the resources associated with a tab, where each tab is contained in a tablist.  <br>


    <br><!-- Spacing line like enter -->


    <b>
       &lt;ul role="tablist"&gt; <br>
        &lt;li role="tab"&gt;</b> Introduction <b>&lt;/li&gt;    <br>
        &lt;li role="tab"&gt;</b> Chapter 1 <b>&lt;/li&gt;   <br>
        &lt;li role="tab"&gt;</b> Chapter 2 <b>&lt;/li&gt;   <br>
       &lt;/li&gt;   <br>
       &lt;div role="tabpanel"&gt; &lt;!----&gt; &lt;/div&gt;  <br>
    </b>


    <br> <!-- Spacing line like enter -->


    <!-- Output code is:-->
```

```
    <b>Then Output code is:</b>
    <ul role="tablist">
        <li role="tab">Introduction</li>
        <li role="tab">Chapter 1</li>
        <li role="tab">Chapter 2</li>
    </ul>
    <div role="tabpanel">
        <!---->
    </div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.49 - role="textbox" -->
<h3> 39.59 - role="textbox"</h3>

<p>
    Input that allows free-form text as its value. <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;textarea title="#" role="textbox"&gt; &lt;/textarea&gt;<br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <textarea title="#" role="textbox"></textarea>
```

```
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.60 - role="timer" -->
<h3> 39.60 - role="timer"</h3>

<p>
    A type of live region containing a numerical counter which indicates an amount of elapsed time from a start point or the time remaining until an end point.     <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;p&gt;   <br>
         &lt;span role="timer"&gt; 60second &lt;/span&gt; <br>
         Second remaining  <br>
        &lt;/p&gt;  <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <p>
        <span role="timer"> 60 second </span>    <br>
        Second remaining
    </p>
</p>

<br> <!-- Spacing line like enter -->
```

```
<!-- Chapter 39.61 - role="toolbar" -->
<h3> 39.61 - role="toolbar"</h3>

<p>
  <b>
    &lt;ul role="toolbar"&gt;  <br>
     &lt;li&gt;&lt;img src="New.png" alt="New"&gt;</b> New <b>&lt;/li&gt;
<br>
     &lt;li&gt;&lt;img src="Open.png" alt="Open"&gt;</b> Open <b>&lt;/li&gt;
<br>
     &lt;li&gt;&lt;img src="Save.png" alt="Save"&gt;</b> Save <b>&lt;/li&gt;
<br>
     &lt;li&gt;&lt;img src="Close.png" alt="close"&gt;</b> Close <b>&lt;/li&gt;
<br>
    &lt;/ul&gt; <br>
  </b>

  <br> <!-- Spacing line like enter -->

  <!-- Output code is: -->
  <b>Then Output code is:</b> <br>
  <ul role="toolbar">
    <li><img src="New.png" alt="New"> New </li>
    <li><img src="Open.png" alt="Open"> Open </li>
    <li><img src="Save.png" alt="Save"> Save </li>
    <li><img src="Close.png" alt="close"> Close </li>
  </ul>
</p>

<br> <!-- Spacing line like enter -->
```

```
<!-- Chapter 39.62 - role="tooltip" -->
<h3> 39.62 - role="tooltip"</h3>

<p>
    A contextual popup that displays a description for an element. <br>

    <br> <!-- Spacing line like enter -->

    <b>
        &lt;pan aria-describedby="slopedesc"&gt;</b> Slope <b>&gt;/span&gt;    <br>
        &lt;div role="tooltip" id="slopedesc"&gt;</b> y=mx+b <b>&lt;/div&gt;   <br>
    </b>

    <br><!-- Spacing line like enter -->

    <!-- Output code is: -->
    <b>Then Output code is:</b> <br>
    <span aria-describedby="slopedesc">Slope</span>
    <div role="tooltip" id="slopedesc">y=mx+b</div>
</p>

<br> <!-- Spacing line like enter -->

<!-- Chapter 39.63 - role="tree" -->
<h3> 39.63 - role="tree"</h3>

<p>
    A type of list that may contain sub-level nested groups that can be collapsed and
expanded.     <br>

    <br> <!-- Spacing line like enter -->
```

```
<!-- Emmet HTML code is: ul[role="tree"]>((li[role="treeitem"]){Part
1}>ul>(li[role="treeitem"]{Chapter})*3)+((li[role="treeitem"]){Part
2}>ul>(li[role="treeitem"]{Chapter})*3)+((li[role="treeitem"]){part
3}>ul>li[role="treeitem"]{Chapter}*3)-->


<b>
  &lt;ul role="tree"&gt;  <br>
   &lt;li role="treeitem"&gt;    <br>
    Part 1  <br>
    &lt;ul role="group"&gt; <br>
    &lt;li role="treeitem"&gt;</b> Chapter 1 <b>&lt;/li&gt;  <br>
     &lt;li role="treeitem"&gt;</b> Chapter 2 <b>&lt;/li&gt;
<br>
     &lt;li role="treeitem"&gt;</b> Chapter 3 <b>&lt;/li&gt;
<br>
    &lt;/ul>   <br>
   &lt;/li&gt;   <br>
   &lt;li role="treeitem"&gt;    <br>
    Part 2  <br>
    &lt;ul role="group"&gt; <br>
    &lt;li role="treeitem"&gt;</b> Chapter 4 <b>&lt;/li&gt;  <br>
     &lt;li role="treeitem"&gt;</b> Chapter 5 <b>&lt;/li&gt;
<br>
     &lt;li role="treeitem"&gt;</b> Chapter 6 <b>&lt;/li&gt;
<br>
    &lt;/ul>   <br>
   &lt;/li&gt;   <br>
   &lt;li role="treeitem"&gt;    <br>
    Part 3  <br>
    &lt;ul role="group"&gt; <br>
    &lt;li role="treeitem"&gt;</b> Chapter 7 <b>&lt;/li&gt;  <br>
```

```
           &lt;li role="treeitem"&gt;</b> Chapter 8 <b>&lt;/li&gt;
<br>
           &lt;li role="treeitem"&gt;</b> Chapter 9 <b>&lt;/li&gt;
<br>
          &lt;/ul>  <br>
         &lt;/li&gt;  <br>
        &lt;/ul&gt; <br>
    </b>

    <br> <!-- Spacing line like enter -->

    <!-- Output code is:: -->
    <b>Then Output code is:</b>  <br>

    <!-- Emmet HTML code is: ul[role="tree"]>((li[role="treeitem"]){Part
1}>ul>(li[role="treeitem"]{Chapter})*3)+((li[role="treeitem"]){Part
2}>ul>(li[role="treeitem"]{Chapter})*3)+((li[role="treeitem"]){part
3}>ul>li[role="treeitem"]{Chapter}*3)-->

    <!-- Just added "li aria-expended="false" " -->
    <ul role="tree" aria-expanded="false">
        <li role="treeitem">
            Part 1
            <ul role="group">
                <li role="treeitem">Chapter</li>
                <li role="treeitem">Chapter</li>
                <li role="treeitem">Chapter</li>
            </ul>
        </li>
        <li role="treeitem"></li>
            Part 2
            <ul role="group">
```

```
            <li role="treeitem">Chapter</li>
            <li role="treeitem">Chapter</li>
            <li role="treeitem">Chapter</li>
        </ul>
    </li>
    <li role="treeitem">
        part 3
        <ul role="group">
            <li role="treeitem">Chapter</li>
            <li role="treeitem">Chapter</li>
            <li role="treeitem">Chapter</li>
        </ul>
    </li>
  </ul>
</p>


<!-- Chapter 39.64 - role="treegrid" -->
<h3> 39.64 - role="treegrid"</h3>




<!-- Chapter 39.65 - role="treeitem" -->
<h3> 39.65 - role="treeitem"</h3>
```

</body>
</html>